

LECTURE 6

Integrity Constraints

CSC270 · Database Systems

COMSATS University Islamabad, Lahore Campus

Agenda

- Domain constraints — keeping individual values legal
- Entity integrity — why no primary key component may ever be NULL
- Referential integrity — keeping foreign keys honest
- General constraints — encoding business rules the model doesn't know about
- How the four constraint categories fit together
- Enforcing referential integrity: CASCADE, RESTRICT, SET NULL

The Extended Schema

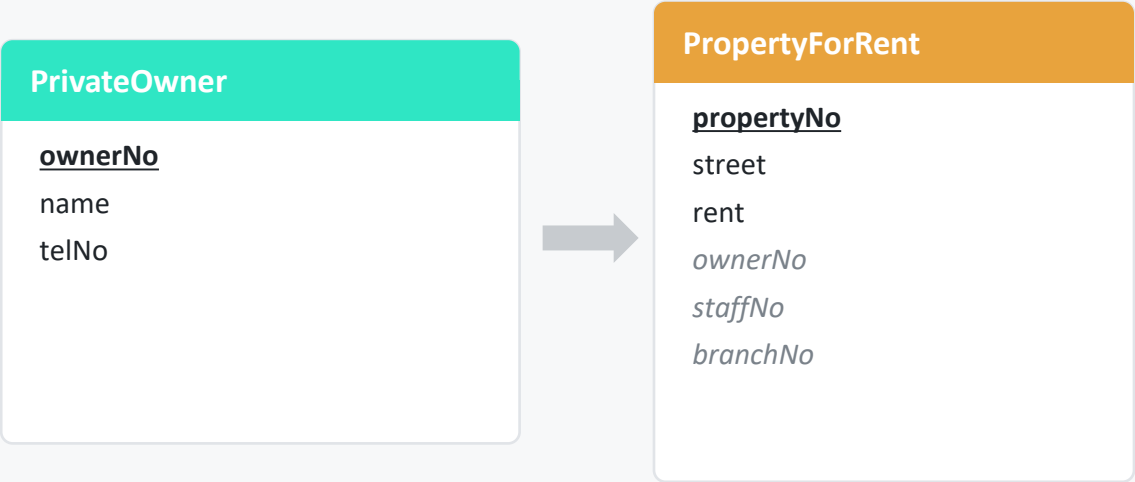
PropertyForRent carries three foreign keys at once — ownerNo, staffNo, branchNo — the perfect relation for testing every integrity rule.

PrivateOwner (ownerNo, name, telNo)

ownerNo	name	telNo
CO46	Joe Keogh	021-234-5678
CO87	Carol Farrel	021-556-7890
CO40	Tina Murphy	042-234-1122
CO93	Tony Shaw	042-556-9988

PropertyForRent — propertyNo, street, city, type, rooms, rent, ownerNo, staffNo, branchNo

propertyNo	type	rent	ownerNo	staffNo	branchNo
PA14	House	22000	CO46	SG14	B003
PL94	Flat	15000	CO87	SA9	B007
PG4	House	35000	CO40	SG14	B003
PG36	Flat	24500	CO93	SG37	B003
PG16	House	45000	CO93	SA9	B007
PG21	Flat	28000	CO87	SL21	B005



Domain Constraints

A domain constraint restricts a single attribute's legal values to its declared domain — the most basic integrity rule, checked in isolation from any other row or table.

```
CREATE TABLE PropertyForRent (  
    propertyNo VARCHAR(5) NOT NULL,  
    rooms      SMALLINT  NOT NULL CHECK (rooms > 0),  
    rent       DECIMAL(8,2) NOT NULL CHECK (rent > 0),  
    type       VARCHAR(5) NOT NULL CHECK (type IN ('House', 'Flat')),  
    ...  
);
```

Domain constraints vs. general constraints

“Rent must be positive” is a domain constraint — one value, one rule. “Rent must be at least 80% of the average rent for that property's city” compares against other rows — that graduates to a general constraint.

Entity Integrity

No attribute participating in a relation's primary key may hold a NULL value.

A primary key's entire job is unique identification. NULL means “unknown” — a row whose identifying value is itself unknown can't reliably be told apart from another row that is also missing it.

Attempted insert into Staff — REJECTED

staffNo	name	position	salary	branchNo
(NULL)	Zara Malik	Assistant	11000	B005

staffNo declared PRIMARY KEY implies NOT NULL automatically — UNIQUE alone still permits NULL.

Composite keys: EVERY component must be non-NULL

In Viewing(clientNo, propertyNo, viewDate, comment), a tuple with a known clientNo but a NULL propertyNo still violates entity integrity — the pair identifies the tuple, and one missing half is enough to break that.

Non-key attributes can still be NULL

Staff.position or PropertyForRent.rooms may legitimately be NULL — the rule is narrowly about the primary key.

Referential Integrity

If a foreign key exists, its value must (1) match a candidate key value currently in the referenced relation, or (2) be wholly NULL.

Attempted insert — REJECTED (SX99 doesn't exist)

propertyNo	ownerNo	staffNo	branchNo
PG55	CO46	SX99	B003

Accepted insert — foreign key wholly NULL

propertyNo	ownerNo	staffNo	branchNo
PG55	CO46	(NULL)	B003

staffNo = NULL doesn't claim any relationship at all — nothing to be inconsistent with. A property may have no staff assigned yet, but never a fabricated one.

Composite foreign keys: wholly NULL, or not at all A foreign key NULL in one column but real in another is itself a violation.

Self-referencing foreign keys

A legal special case: Branch(branchNo, street, city, postcode, mgrStaffNo) could hold mgrStaffNo as a foreign key into Staff — and Staff already references Branch. A relation can even reference itself, e.g. Staff.supervisorStaffNo → Staff.staffNo.

General Constraints

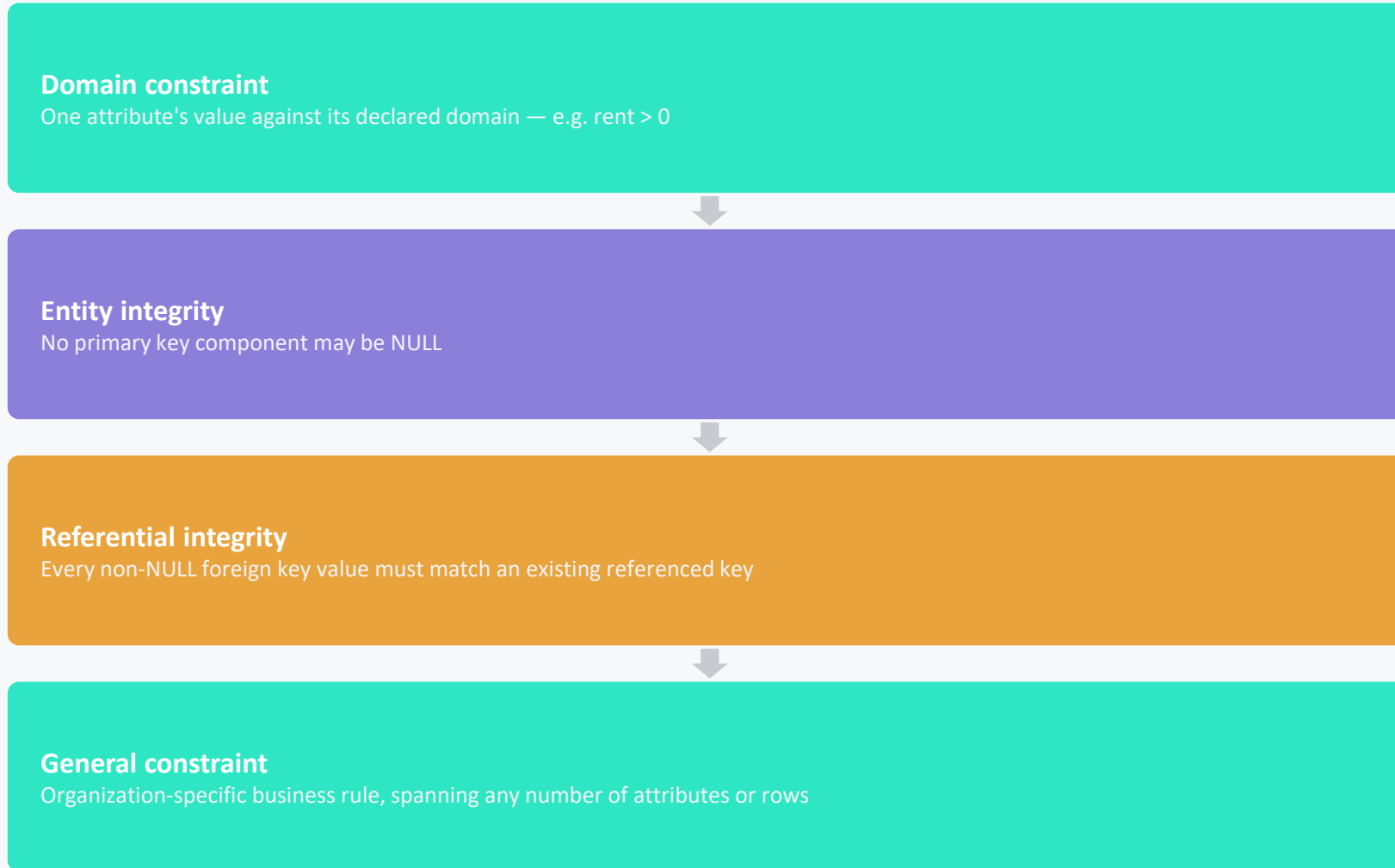
Organization-specific business rules that go beyond domain, entity, and referential integrity — the relational model has no built-in vocabulary for them.

- No member of staff can manage more than 100 properties at a time.
- A manager's salary must exceed every assistant's salary at the same branch.
- A property's rent must fall between PKR 5,000 and PKR 100,000.
- Staff can't handle a viewing for a property they themselves manage.

```
ALTER TABLE PropertyForRent
ADD CONSTRAINT chk_rent_range
CHECK (rent BETWEEN 5000 AND 100000);
```

Simple, single-table CHECKs are easy for the DBMS to enforce directly. Rules comparing across many rows (“manager > every assistant”) usually need triggers, stored procedures, or application logic — but they remain integrity constraints either way.

Integrity Constraints, Together



All four exist for the same reason: a database is only trustworthy if the DBMS refused every write that would have made a query result false.

Enforcing Referential Integrity

Domain and entity integrity: the offending INSERT/UPDATE is simply rejected. Referential integrity is more interesting — a violation can come from either side, so the DBMS needs a policy per direction.

Deleting Staff row SG14 (who manages PA14 and PG4) — three referential actions:

RESTRICT

Refuse the delete while PA14/PG4 still reference SG14

CASCADE

Delete SG14, then auto-delete PA14 and PG4 too

SET NULL

Delete SG14, set staffNo to NULL on PA14 and PG4

```
CREATE TABLE PropertyForRent (  
    ...  
    staffNo    VARCHAR(5),  
    branchNo   VARCHAR(4) NOT NULL,  
    FOREIGN KEY (staffNo) REFERENCES Staff(staffNo) ON DELETE SET NULL,  
    FOREIGN KEY (branchNo) REFERENCES Branch(branchNo) ON DELETE RESTRICT  
);
```

Choosing a Referential Action

This is a design decision, not a default Ask: “if the referenced row disappears, does the referencing row still make sense on its own?”

Relationship	If the referenced row disappears...	Favor
Viewing → Client	A viewing record makes no sense without the client	CASCADE / RESTRICT
PropertyForRent → staffNo	The property still makes sense, just unassigned	SET NULL
PropertyForRent → branchNo	A property cannot exist without belonging to some branch	RESTRICT

Getting this wrong is a common, expensive real-world bug — either silent data loss from an over-eager CASCADE, or a wall of RESTRICT errors when SET NULL would have been sensible.

KEY TAKEAWAYS

Integrity Constraints

- Domain constraints restrict a single attribute's value to its declared, legal domain.
- Entity integrity: no attribute that is part of a primary key — single or composite — may ever be NULL.
- Referential integrity: every non-NULL foreign key value must match an existing candidate key value, or be wholly NULL.
- General constraints capture business rules the relational model has no built-in vocabulary for.
- Referential integrity enforcement differs by direction: referencing-side writes that would break it are always rejected; referenced-side deletes/updates offer a policy — RESTRICT, CASCADE, or SET NULL — chosen by what the relationship means.